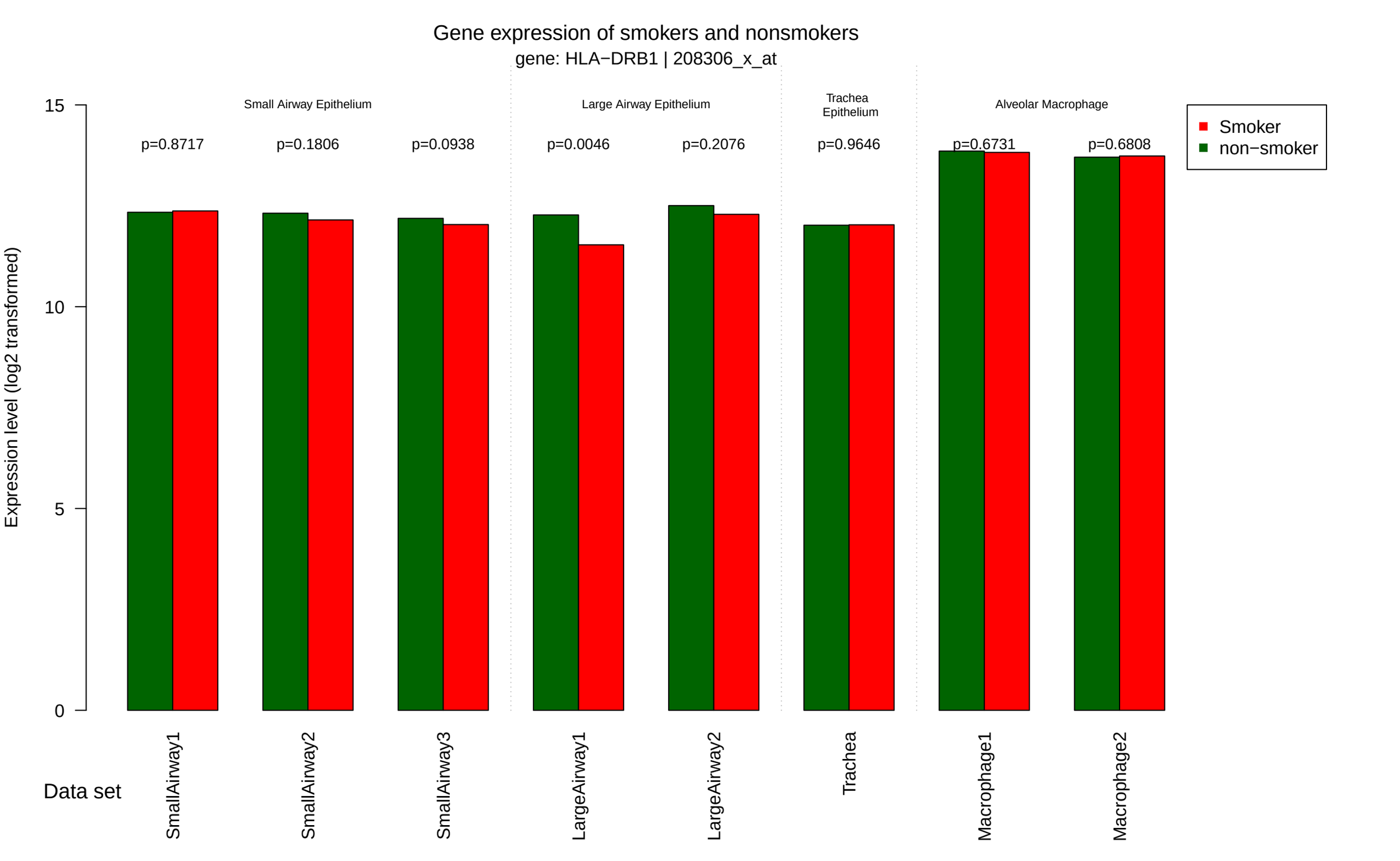
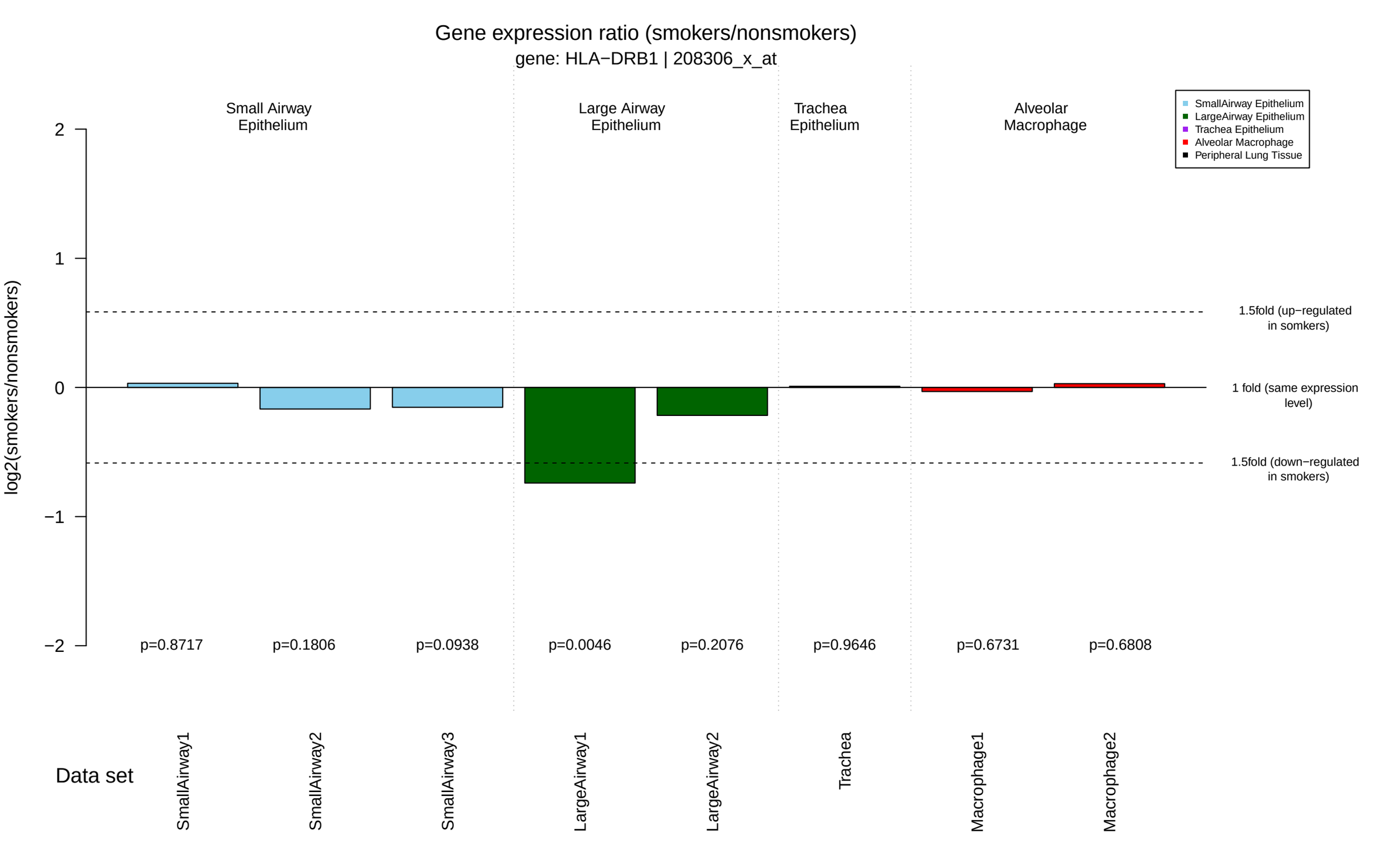
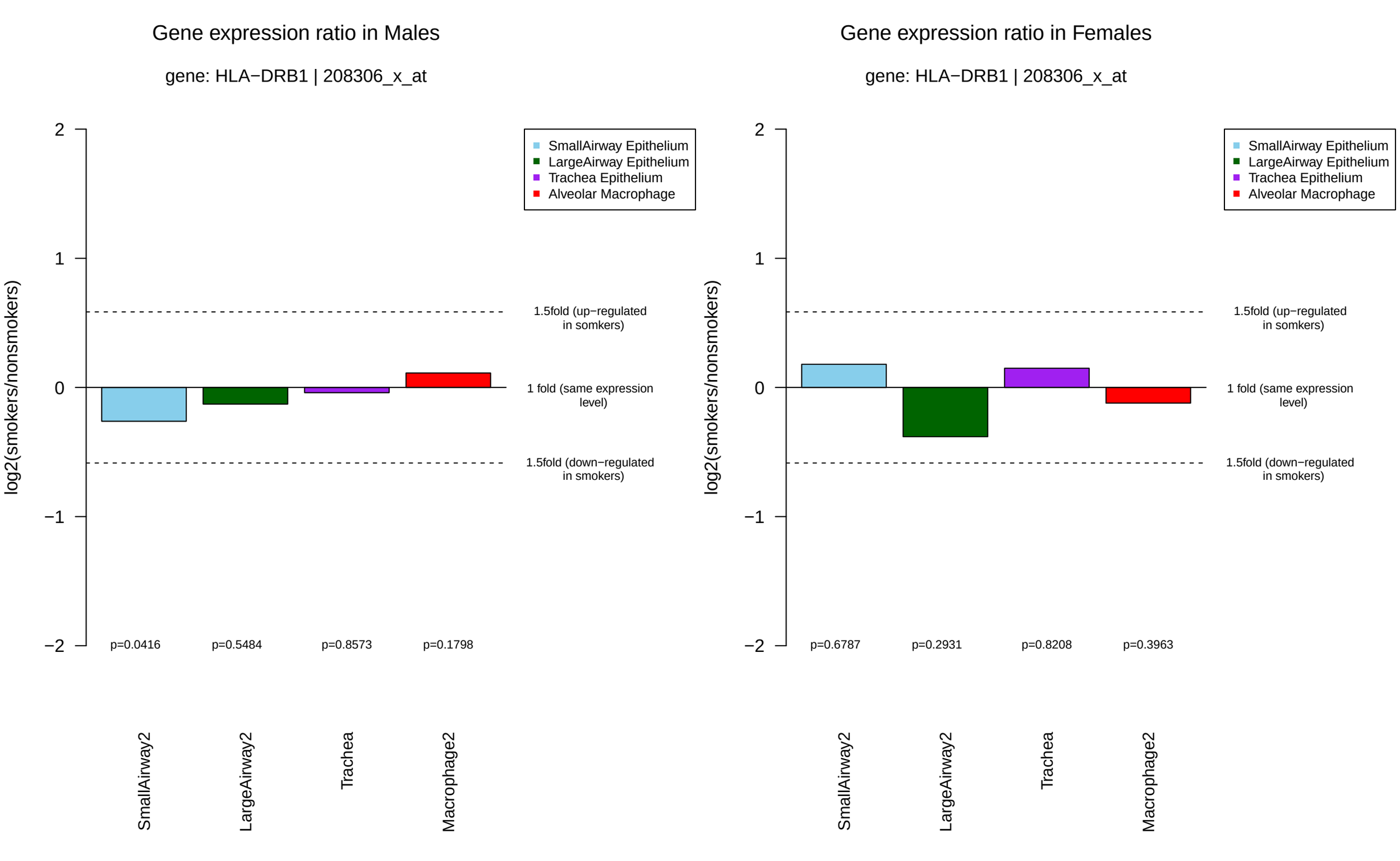
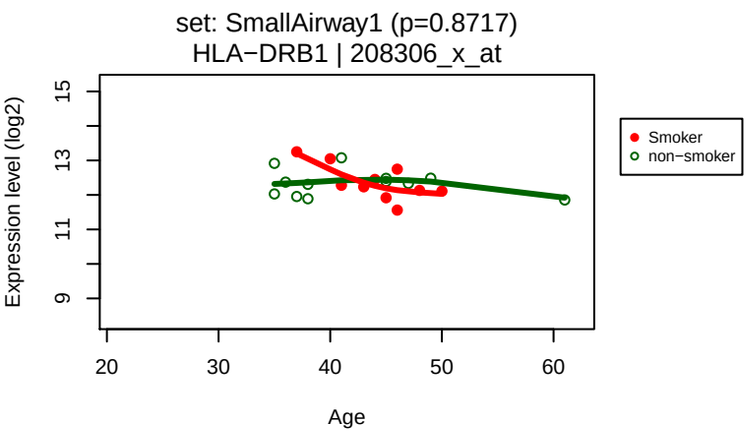
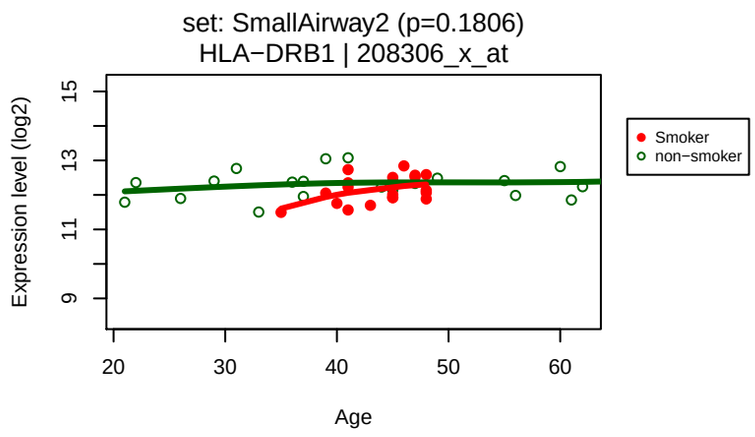
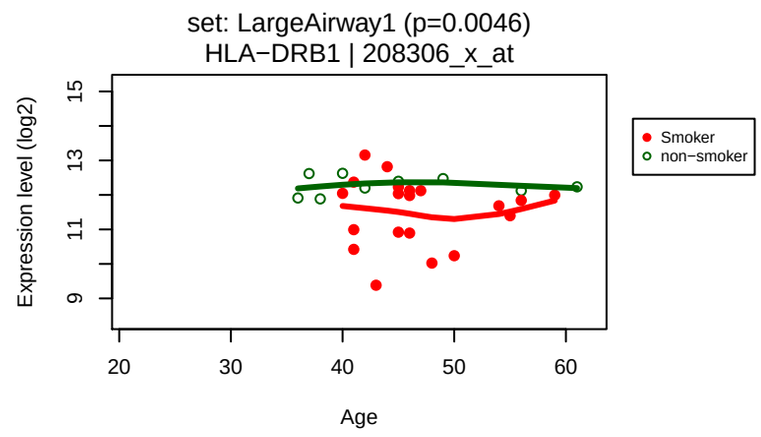

Gene expression of smokers and nonsmokers
gene: HLA−DRB1 | 208306_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8717
p=0.1806
p=0.0938
p=0.0046
p=0.2076
p=0.9646
p=0.6731
p=0.6808
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

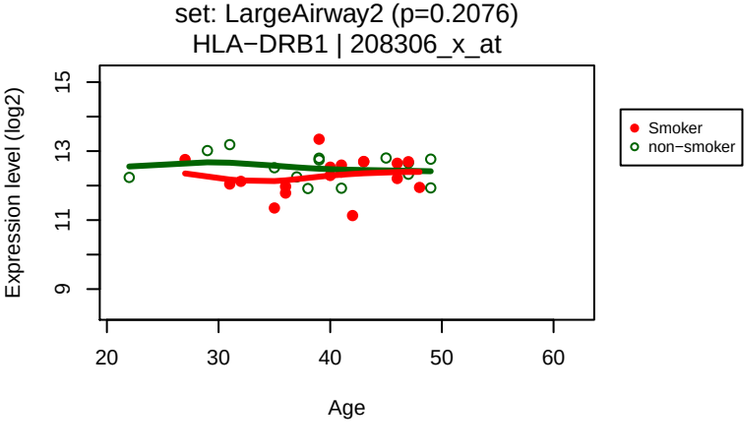

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB1 | 208306_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8717
p=0.1806
p=0.0938
p=0.0046
p=0.2076
p=0.9646
p=0.6731
p=0.6808
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

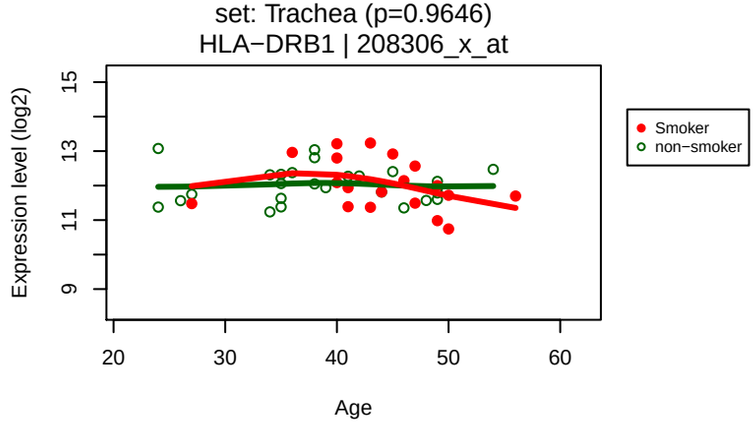

Gene expression ratio in Males
gene: HLA−DRB1 | 208306_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0416
p=0.5484
p=0.8573
p=0.1798
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB1 | 208306_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6787
p=0.2931
p=0.8208
p=0.3963
SmallAirway2
LargeAirway2
Trachea
Macrophage2

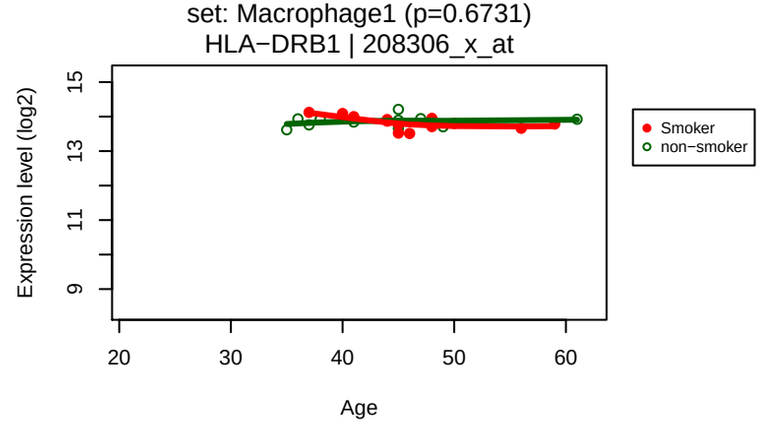

set: SmallAirway1 (p=0.8717)
HLA−DRB1 | 208306_x_at
Expression level (log2)
Age
Smoker
non−smoker

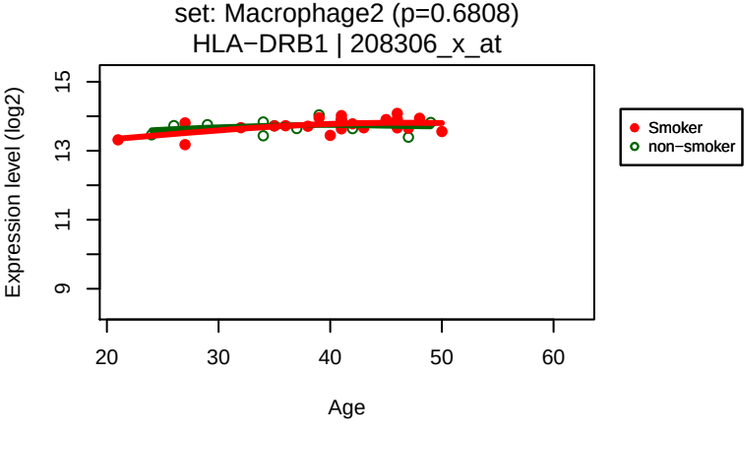

set: SmallAirway2 (p=0.1806)
HLA−DRB1 | 208306_x_at
Expression level (log2)
Age
Smoker
non−smoker

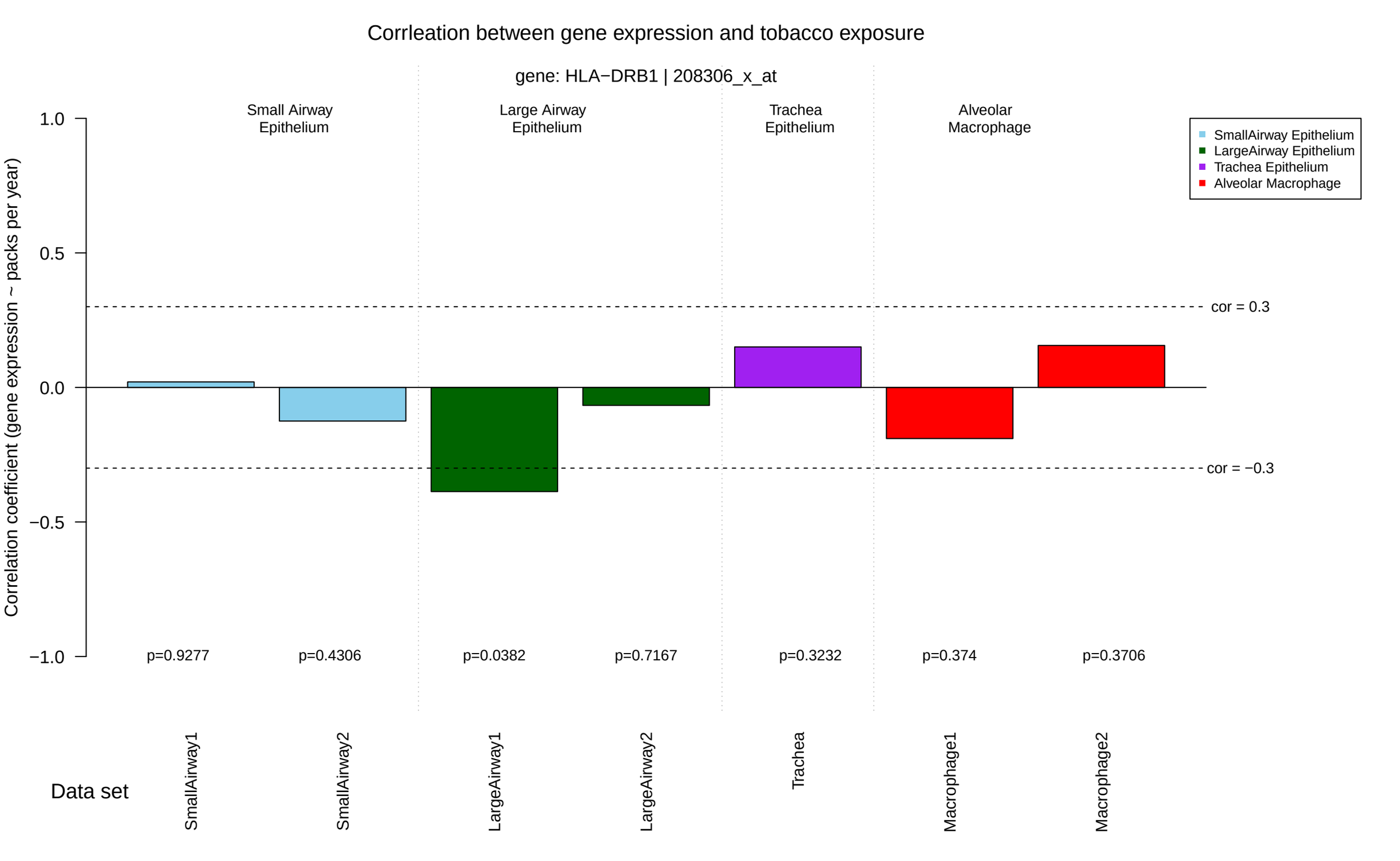

set: LargeAirway1 (p=0.0046)
HLA−DRB1 | 208306_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2076)
HLA−DRB1 | 208306_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9646)
HLA−DRB1 | 208306_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6731)
HLA−DRB1 | 208306_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6808)
HLA−DRB1 | 208306_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB1 | 208306_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9277
p=0.4306
p=0.0382
p=0.7167
p=0.3232
p=0.374
p=0.3706
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2